



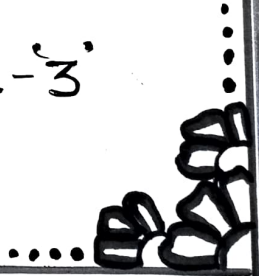
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C02 - ASSESSMENT TOOL - '3'



4. Footprinting Tools - Conducting Competitive Intelligence

A cybersecurity consultant has been assigned to perform a passive security assessment for ABC Technologies. The consultant must gather only publicly available information [OSINT] without directly communicating with the company's servers.

Objective -

- Domain registration details.
- Employee email addresses.
- Technology Stack
- Publicly exposed Subdomains.

1. Domain Registration Details -

• Tool used : WHOIS

WHOIS provides registration information about a domain name.

Command -

whois abctechnologies.com

Information Collected

- Domain owner
- Register name
- Register date
- Expiration date
- Name Servers
- Administrative Contact
- Technical contact

Example Output -

Domain :

abctechnologies.com

Register :

GoDaddy

Created :

12-Jan-2021

Expires :

12-Jan-2028

Name Servers :

ns1.example.com

ns2.example.com

Importance in Competitive Intelligence

WHOIS helps identify:

- Domain ownership
- Organization responsible
- Hosting provider
- DNS infrastructure
- Domain age
- possible acquisitions or mergers.

2. Employee Email Addresses

Tool used : theHarvester

TheHarvester collects publicly available email addresses

- Google
- Bing
- Yahoo
- LinkedIn
- PGPServers
- Github

Command -

theHarvester -d abctechnologies -b all

Information collected -

→ Employee emails

- Hostnames
- public IP addresses
- Related domains

Example output -

admin@abctechnologies.com
hr@abctechnologies.com
support@abctechnologies.com
john.smith@abctechnologies.com

Importance in competitive Intelligence

Email addresses reveal :

- Employee naming conventions
- Department structure
- Executive contacts
- possible phishing targets

3. Technology stack Identification

Tool used : Wapplayzer

Wapplayzer detects technologies used by websites.

Information Identified

- Web sever
- CMS
- Javascript frameworks
- Analytics
- programming language
- Database
- CDN

Example result -

Operating System :

Linux

Web Sever :

Apache

Language :
PHP

Framework :
Larvel

cms :
Wordpass

Analytics :
Google analytics

CDN :
Cloudflare

Importance in competitive Intelligence

Knowing the technology stack helps identify :

- Outdated software
- Known vulnerabilities
- Development platforms
- Third-party services
- Security mechanisms

Summary of footprinting Tools

Tool	Purpose	Information Gathered
WHOIS	Domain register	Owner, registrar
theHarvester	Email enumeration	Employee email, hosts
Wappalazer	Technology detection	CMS, sever, framework
Sublist3r	Subdomain discovery	Public subdomains

Benefits of competitive Intelligence -

- Understands external attack surface.
- Identities exposed services.
- Detects technology used.
- Helps perform risk assessment.

2. DNS Zone Transfer

DNS zone Transfer are permitted and explain how the retrieved information can assist during reconnaissance.

Command -

```
dig AXFR @ns1.abctechnologies.com  
abctechnologies.com
```

[or]

```
dig @ns1.abctechnologies.com  
AXFR abctechnologies.com
```

Successful Zone Transfer Output

```
www.abctechnologies.com  
mail.abctechnologies.com  
VPN.abctechnologies.com  
ftp.abctechnologies.com  
dev.abctechnologies.com  
api.abctechnologies.com  
db.abctechnologies.com  
ns1.abctechnologies.com
```

Reconnaissance Benefits

If a zone transfer succeeds, an attacker gains:

- Complete DNS Structure
- Internal naming conventions
- Server roles
- Hidden Systems
- VPN locations
- Mail infrastructure
- Backup servers
- Development Servers.

Alternative tools

nslookup

Sever ns1.abctechologies.com
ls -d abctechologies.com

Security Recommendations

- Disable public AXFR requests
- permit zone transfer only between authorized DNS Sever
- Restrict DNS logging and monitoring
- Regularly audit DNS configurations.
- Enable DNS logging and monitoring.

DNS Zone transfer are essential for DNS replication and become a sever security risk when configured.

Verifying AXFR using tools like dig helps identify their weakness.

Restricting zone transfers protects sensitive infrastructure information from authorized.

3. Social Engineering Attacks and countermeasures

- This is phishing attack, a common from a social engineering.
- Phishing tricks users into revealing sensitive information by impersonating trusted entities.
 - Fake senders address
 - Urgent language
 - fear a content suspension
 - Fake login pages
 - Malicious links
 - Company branding

Counter measures

1. Verify the Sender

Employee should carefully inspect the Sender's email.

Example :

it-Support@abc-tech-security.com

instead of

it@abctechnologies.com

2. Avoid clicking Unknown Links

- check whether the destination matches the organisation Official domain.
- This prevents visiting fraudulent websites.

3. Verify through Official channels.

- contact the IT department using Official contact information.
- Do not reply solely on the email's instructions.

4. Never share credentials

- legitimate IT departments do not request passwords through email.
- Employee should never enter credentials on links received via email without verification.

Report Suspicious Emails

forward suspicious emails to the Organizations of IT's or Security team.

- Block malicious domains.
- Warn other employees.
- Prevent widespread compromise.

Participate in Security Awareness Security

Regular cybersecurity training helps employees.

- Recognize phishing emails.
- Identify suspicious requests.
- Respond appropriately to potential attacks

Actions prevent Credential Theft

Verify sender → Detects spoofed emails

Avoid unknown links → prevents access to fake login.
contact IT directly → confirms legitimacy
Never share passwords → Stops credential disclosure.
Enable MFA → Reduces impact of stolen passwords.
Security awareness training → Improves user vigilance.

Conclusion -

The email described is a phishing attack designed to steal employee credentials. By verifying the sender, avoiding suspicious links, using official communication channels enabling MFA reduce the risk of credential theft.